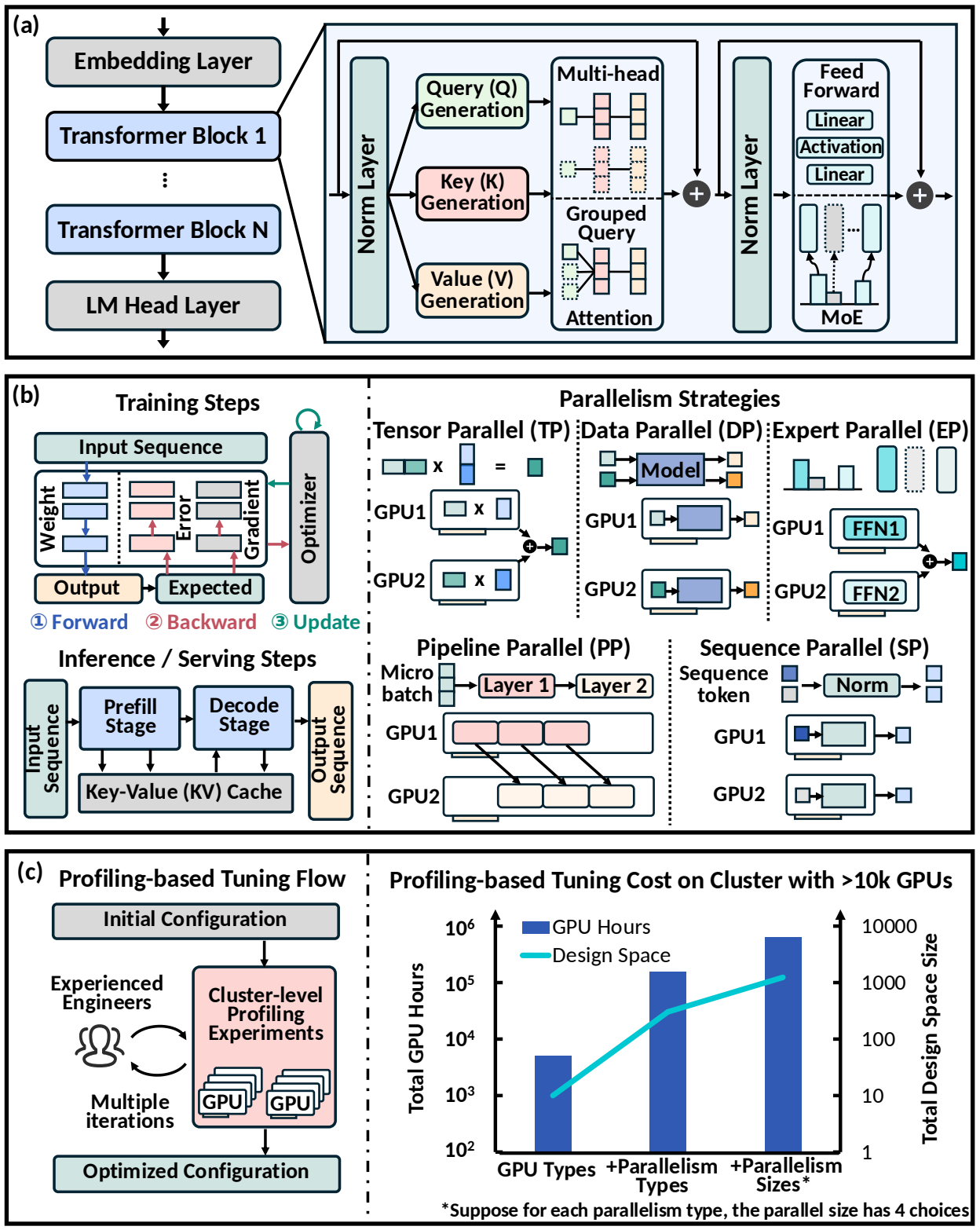

(a)
Embedding Layer
Transformer Block 1
Transformer Block N
LM Head Layer
Norm Layer
Query (Q) Generation
Key (K) Generation
Value (V) Generation
Multi-head
Grouped Query
Attention
Norm Layer
Feed Forward
Linear
Activation
Linear
MoE
(b)
Training Steps
Input Sequence
Weight
Error
Gradient
Optimizer
Output
Expected
① Forward
② Backward
③ Update
Inference / Serving Steps
Input Sequence
Prefill Stage
Decode Stage
Output Sequence
Key-Value (KV) Cache
Parallelism Strategies
Tensor Parallel (TP)
Data Parallel (DP)
Expert Parallel (EP)
GPU1
GPU2
Model
FFN1
FFN2
Pipeline Parallel (PP)
Micro batch
Layer 1
Layer 2
Sequence Parallel (SP)
Sequence token
Norm
(c)
Profiling-based Tuning Flow
Initial Configuration
Experienced Engineers
Cluster-level Profiling Experiments
GPU
Multiple iterations
Optimized Configuration
Profiling-based Tuning Cost on Cluster with >10k GPUs
GPU Hours
Design Space
Total GPU Hours
Total Design Space Size
GPU Types
+Parallelism Types
+Parallelism Sizes*
*Suppose for each parallelism type, the parallel size has 4 choices